Delivery Mechanism and Community Usage Metrics

Delivery Mechanisms

Hosted service. CryoStack will be delivered through the Georgia Tech gateway at https://cryostack.eas.gatech.edu. The gateway will provide documentation, authenticated workspaces, saved configurations, experiment tracking, the Frozen Legacies historical-radar catalog, LIVIST (Living Ice Sheet Temperature) exploration using radar- and borehole-derived products, CryoLauncher model execution, and ICESEE data assimilation. Core laptop-scale tutorials will remain usable without remote HPC.

Open software and environments. Source, issues, releases, adapter templates, and governance will be public in the ICESEE-project GitHub organization. Twice-yearly tagged releases will be archived with DOI records. Supported execution environments will be delivered as source recipes, Spack locks, and OCI/Apptainer images pinned by digest, with software bills of materials, license manifests, compatibility matrices, and validation reports.

Remote execution. Typed, allow-listed execution adapters will stage declared inputs, submit and monitor jobs, and recover declared outputs on qualified backends. Georgia Tech PACE is the reference backend; AWS Batch (Fargate by default, EC2 On-Demand as an opt-in profile) provides a portable cloud profile for project-managed or user-managed accounts, allocations, and credits; CryoStack does not require AWS, and outside budgeted project activities AWS bills the researcher's own account. A bounded access pilot, if approved, gives up to 50 participants a small introductory allocation for eligible CryoLauncher and ICESEE workflows. The application-level experiment specification will remain consistent across profiles. Credentials remain with the user connector or an approved credential service, not in the public gateway database.

Community delivery. The project will offer one annual hands-on workshop, quarterly office hours in Years 2–3, recorded/literate tutorials, and a public support channel. A contribution guide, adapter proposal template, code of conduct, roadmap, support policy, and governance process will make the service usable and extensible beyond the originating team.

Yearly Targets and Collection

Targets are evaluation commitments, not claims of present adoption. "Unique users" are privacy-preserving counts of authenticated people who complete at least one scientific run in the reporting year. "External group" means a non-project research or teaching group with at least two completed runs or a documented curricular/research use. Automated traffic and project-team tests are excluded from adoption counts.

Quality, Science, and Broader-Impact Evidence

Each production adapter must pass schema/API tests; deterministic restart and seeded reproducibility tests; a model-specific scientific benchmark; an end-to-end assimilation test with predefined error/spread tolerances; and a cross-backend scientific-equivalence test on every qualified backend it claims to support, comparing field norms, integrated or conserved quantities, ensemble statistics, or DA-specific diagnostics against versioned tolerance envelopes rather than requiring matching runtime or cost. The PISM adapter must complete a synthetic-truth experiment and a documented observational demonstration. Benchmark reports will record code/environment versions, runtime, resource use, and tolerances.

Workshop reporting will include attendance, career stage, institution type, completion of the hands-on workflow, and an anonymous usefulness survey. Demographic information will be optional,

Metric	Y1	Y2	Y3	Collection/evidence
Qualified model adapters (cumulative)	4	5	5	CryoBench registry and release tags; PISM is the fifth
Qualified dataset adapters (cumulative)	1	2	2	Validated manifests, provenance/QC reports, and release tags
Adapters passing required release gates	80%	100%	100%	Signed CI/benchmark reports
Qualified remote backend profiles (cumulative)	1	2	2	PACE and AWS qualification checklists
PISM-specific changes confined to the adapter/configuration layer (shared-code changes only via documented extension interfaces)	–	$\geq$90%	$\geq$90%	Code-review diff classification at production release
Tagged stack releases per year	2	2	2	GitHub/Zenodo release records
Core-workflow availability, excluding planned maintenance	95%	97%	97%	External uptime monitor (Year-1 instrument; current checks are internal) and incident log
Median time to acknowledge high-severity service issues	2 d	1 d	1 d	Public/private issue timestamps
Annual unique scientific users	25	60	100	De-identified account/run records
Completed non-test scientific runs per year	150	500	1,000	Experiment registry; bot/team tests excluded
External adopting groups (cumulative)	2	5	8	Opt-in affiliation plus qualifying usage/interview
Annual workshop participants	25	40	50	Registration and attendance records
Core user-task completion rate, each of the four applications	65%	75%	80%	Application-specific task studies
Access-pilot first-run completion (if pilot approved)	–	70%	80%	Pilot run records; cost per completed workflow
Allocation enforcement tests passed (attribution, refusal, pool isolation, recovery)	–	100%	100%	Task 2.4 test reports; reconciled estimate-vs-actual cost
Median System Usability Scale score	65	70	75	Anonymous post-study survey
External contributors (cumulative)	1	2	3	Accepted code, docs, test, or adapter contributions
Research/teaching outputs citing CryoStack (cumulative)	1	3	6	User survey, DOI/citation alerts, manual verification

minimized, reported only in aggregate, and suppressed for small cells. Accessibility checks and user-task studies will generate issue lists and closure evidence. Annual public reports will compare targets with actuals, explain shortfalls, identify corrective actions, and publish machine-readable aggregates where privacy permits.

Instrumentation and Responsibility

The durable identity/experiment registry and expanded Control Center are the sources for users, runs, backends, status, audit events, availability, and reproducibility fields; they will store no scientific credentials. GitHub and Zenodo APIs provide release/contribution evidence. An external uptime monitor and incident log provide operations data. Registration forms, task-study instruments, and short annual opt-in user surveys provide community evidence. The PI reviews technical and operations metrics monthly; the Co-PI reviews scientific and user metrics quarterly. Frozen yearly extracts, definitions, exclusion rules, and analysis scripts will be archived with each annual report so that trends remain auditable if instrumentation changes.